INTERNAL SPECIFICATION FOR **CERAMIC TILES**

Sr. No	CHARACTERISTICS	(Rectified) 60.0x60.0x1.0 cm		(Rectified) 50.0x50.0x1.4 cm	
1	Nominal size **(mm)**	600.0x600.0x1.0		500.0x500.0x14	
2	Work size **(mm)**	600.0x600.0x1.0		500.0x500.0x14	
		ISO 13006	**SCC. spec.**	**ISO 13006**	**SCC. spec.**
3.1	Size variation from work size **(%)**	± 0.30	± 0.30	± 0.30	± 0.30
3.2	In **mm**	± 1.00	± 1.00	± 1.00	± 1.00
3.3	Size range in **mm**		599.9 – 600.4		499.9 - 500.4
3.4	Max.devation in diagonal (mm)		0.8		0.8
4	Thickness **(mm)**		10		14
4.1	Thickness deviation **(%)**	± 10.0	± 5.0	± 10.0	± 5.0
4.2	Thickness Range (mm)		9.50 – 10.50		13.5 - 14.5
5	Centre Curvature	mm	mm	mm	mm
5.1	a) Convex **(+)**	+ 2.00	+ 1.80	+ 1.80	+ 1.80
5.2	b) Concave **(-)**	- 2.00	- 0.30	- 1.80	- 0.30
6	Warpage	± 2.00	± 1.80	± 1.80	± 1.60
7	Edge Curvature				
7.1	a) Convex **(+)**	+ 2.00	+ 1.50	+ 1.80	+ 1.40
7.2	b) Concave **(-)**	- 2.00	- 0.40	- 1.80	- 0.40
8	Max deviation from Straightness **(mm)**	± 0.80	± 0.50	± 0.80	± 0.50
9	Max deviation from Rectangularity **(mm)**	± 1.50	± 1.50	± 1.50	± 1.40
10	Water Absorption **(%)**	> 10	16 ± 2	> 10	16 ± 2
11	Coefficient to Linear thermal expansion **($x10^{-6}K^{-1}$)**	-	≤ 8	-	≤ 8
12	Resistance to surface abrasion **(Class)**	Manufacturer to state class	2 - 5	Manufacturer to state class	3 - 5
13	Breaking strength **(N)**	≥600	≥800	≥600	≥800
14	Modulus of Rupture **MPa (Kg/cm^2)**	Min. 12 (120)	Min. 15 (150)	Min. 12 (120)	Min. 15 (150)
15	Crazing resistance **(5bar/2 hr cycle)**	Required	Resistant	Required	Resistant
16	Scratch resistance Moh's Scale (Glossy)	-	≥3	-	≥3
17	Scratch resistance Moh's Scale (Matte)	-	≥6	-	≥6
18	Staining resistance **(Class)**	Min. 3	Min. 3	Min. 3	Min. 3
19	Chemical resistance to Acids and alkalis **(Class)**	Required if agreed	B	Required if agreed	B
20	Resistance to household Chemicals **(Class)**	Min. B	B	Min. B	B